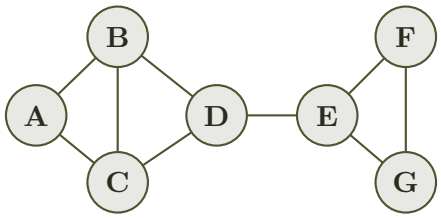
Module 8: Graph Theory and Networks — Summary

Central idea: The arrangement of connections determines who can reach whom and whether a system can continue working after a failure.



Represent a network

A **node** (vertex) is a place or organization. An **edge** is a direct connection. The county network has nodes A–G and nine two-way links: AB, AC, BC, BD, CD, DE, EF, EG, FG. The diagram is not a map to scale: nearby dots are not automatically connected.

Count direct neighbors

The **degree** of D is 3 because D connects to B, C, E. The degrees of A–G are 2, 3, 3, 3, 3, 2, 2; their sum is 18. Each of the nine edges is counted at two ends. This is the **handshake rule**: sum of degrees = $2 \times$ number of edges.

Paths and distance

A **path** follows existing edges. A–B–D–E–G uses **four edges**, so its length is 4. A **shortest path** uses the fewest edges in an unweighted network. A and G can reach each other even without a direct A–G link. When paths represent travel, we may attach times to the links and add minutes instead.

Connected components and bridges

The original network has one **connected component**: all locations can reach one another. If D–E fails, there are two: $\{A, B, C, D\}$ and $\{E, F, G\}$. D–E is a **bridge**, because losing it separates the network. B–C is not a bridge; B can still reach C through A or D.

Module 8: Graph Theory and Networks — Summary (continued)

Use the mathematics to compare choices.

One new link, different outcomes

Suppose the county can add either A–D or D–F. Both give ten links. A–D makes it easier for A to reach the school directly, but D–E remains a bridge. D–F creates an alternate route between regions. If D–E fails, A–B–D–F–G still works. The **arrangement** of links matters more than their number alone.

An alert moves one link per round

Begin with A informed in round 0. In each round, informed locations send to direct neighbors. Newly informed locations relay in the *next* round; every available link works.

Round	Newly informed	Total informed
0	A	A
1	B, C	A, B, C
2	D	A, B, C, D
3	E	A, B, C, D, E
4	F, G	all seven

Under this rule, the first arrival round equals a location's shortest-path distance from A. A failed bridge can prevent delivery to some locations entirely.

A five-step method

1. Say what nodes and edges mean in the situation.
2. Decide whether the question is about direct contacts, routes, failure, or spreading.
3. Count neighbors or trace paths; verify every link exists.
4. Compare the original network with one specific change.
5. Explain a limitation (for example travel time, unreliable contacts, or accessibility).

Remember: Two networks can have the same number of edges and very different routes and vulnerabilities. The arrangement matters.